Principles of Quantum Mechanics

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1 Introduction

Recall from IB Quantum Mechanics we have the following:

(i) Wave-particle duality. Waves behave like particles, like light quanta, and vice versa (particles behave like waves).

(ii) We have a wavefunction $\psi(x)$ for a particle which we interpret as a probability density $|\psi(x)|^2$ (i.e. probability is intrinsic to the theory).

(iii) Observables become operators on wavefunctions:

- $\hat{x} \to x$;
- $\hat{p} \to -i\hbar\frac{\partial}{\partial x}$

The lack of commutation, since $[\hat{x}, \hat{p}] = i\hbar$, leads to the uncertainty relation.

(iv) Schrödinger's equation with Hamiltonian H specifies dynamics, and eigenvalues of H are the allowed energy levels of the system.

(v) All of this explains the hydrogen atom structure and the resulting line spectra.

The aims of this course are the following:

(i) Reformulate quantum mechanics in a more powerful, flexible and useful form: the Dirac formulation, which goes beyond wavefunctions. The previous description appears as a particular realisation of our generalisation of quantum states and also leads to alternatives such as

$$\tilde{\psi}(p) \quad \text{with} \quad \hat{x} \to i\hbar\frac{\partial}{\partial p} \quad \text{and} \quad \hat{p} \to p.$$

(ii) This more general approach allows much simpler analysis of known problems (for example, the Simple Harmonic Oscillator), but also lots of new things such as *spin* of particles and symmetries (translations, rotations) and the related conservation laws.

(iii) We will also look at identical particles (bosons and fermions).

(iv) We will develop a framework for quantising more general systems such as EM fields.

(v) We will also set up perturbation theory, which is a practical tool for computation, but we will not heavily consider the detailed applications.

2 Dirac Formalism

2.1 Concepts and Definitions

A quantum system is described at each instant by a *state* $|\psi\rangle$ which belongs to a complex vector space V; any non-zero element of this space V is a possible state. Thus

$$|\psi\rangle, |\phi\rangle \in V \implies \alpha|\psi\rangle + \beta|\phi\rangle \in V \quad \text{for any } \alpha, \beta \in \mathbb{C}.$$

Physically, this is a superposition principle leading to wave-like behaviour, but these states are *not wavefunctions*. There are also *dual* or conjugate states $\langle\psi|$ which belong to the dual space

$V^\dagger$. By definition, states and duals can be combined or paired to give a complex number. The map

$$(\langle\psi|\,, |\phi\rangle) \mapsto \langle\psi|\phi\rangle$$

defines a map from $V^\dagger \times V \to \mathbb{C}$ with

$$\langle\phi|\,(\alpha_1\,|\psi_1\rangle + \alpha_2\,|\psi_2\rangle) = \alpha_1\,\langle\phi|\psi_1\rangle + \alpha_2\,\langle\phi|\psi_2\rangle$$

and

$$(\beta_1\,\langle\phi_1| + \beta_2\,\langle\phi_2|)\,|\psi\rangle = \beta_1\,\langle\phi_1|\psi\rangle + \beta_2\,\langle\phi_2|\psi\rangle\,.$$

The spaces V and $V^\dagger$ come with an inner product which can be described as a one-to-one correspondence between the spaces

$$\begin{aligned} V &\to V^\dagger \\ |\psi\rangle &\to \langle\psi| = (|\psi\rangle)^\dagger\,. \end{aligned}$$

We also have that if we take a linear combination of states to the dual space, the coefficients have to have their complex conjugates taken when mapping from the dual space to V or vice versa. Hence the inner product is

$$|\phi\rangle\,, |\psi\rangle \mapsto \langle\phi|\psi\rangle = (|\phi\rangle)^\dagger\,|\psi\rangle$$

and it is assumed to obey

$$\langle\phi|\psi\rangle = (\langle\psi|\phi\rangle)^*$$

since the inner product is Hermitian, and the inner product defines a positive-definite norm on V:

$$\|\,|\psi\rangle\,\|^2 = \langle\psi|\psi\rangle\,.$$

Remark. Knowing $\langle\phi|\psi\rangle$ for all $\langle\phi|$ determines $|\psi\rangle$ and vice versa.

We can show this by

$$\langle\phi|\psi_1\rangle = \langle\phi|\psi_2\rangle \implies \langle\phi|\chi\rangle = 0,$$

where $|\chi\rangle = |\psi_1\rangle - |\psi_2\rangle$. If this holds for all $\langle\phi|$, we can set $\langle\phi| = \langle\chi|$, so by positive definiteness we have that the norm of $|\chi\rangle$ is zero, and therefore $|\chi\rangle = 0$ too.

We also have that the physical content of any state is unchanged by

$$|\psi\rangle \to \alpha\,|\psi\rangle$$

for any non-zero $\alpha \in \mathbb{C}$. So we will usually normalise to make the state have norm 1, but we can still change it by a phase factor.

Remark. Relative phases can be important in expressions where we're adding a linear combination of elements of V. We also assume the space V is complete so appropriate sequences and series converge in V.

A complete inner product space is called a *Hilbert space*.

Definition. (Operator) An *operator* Q is a linear map on states

$$|\psi\rangle \to Q\,|\psi\rangle\,.$$

The same operator can be thought of as acting to the left on dual states, so

$$\langle\phi| \to \langle\phi|\,Q$$

where the right-hand side is defined by

$$(\langle\phi|\,Q)\,|\psi\rangle = \langle\phi|\,(Q\,|\psi\rangle) = \langle\phi|\,Q\,|\psi\rangle\,.$$

Remark. States and operators $|\psi\rangle, \langle\phi|, Q$ can be dealt with by rulers that mirror those for complex vectors $\mathbf{u}(\text{col}), \mathbf{v}^\dagger(\text{row})$ and a matrix A. So we can compare $\langle\psi|\,Q\,|\phi\rangle$ with $\mathbf{v}^\dagger A\mathbf{u}$ like we do with inner products.

Definition. (Eigenstate) For any operator Q with say that $|\psi\rangle$ non-zero is an *eigenstate* with *eigenvalue* λ if

$$Q\,|\psi\rangle = \lambda\,|\psi\rangle$$

or equivalently

$$\langle\psi|\,Q^\dagger = \lambda^*\,\langle\psi|\,.$$

Remark. For a general Q, λ may be complex.

Definition. (Hermitian Conjugate) For any operator Q, the *Hermitian conjugate* or *adjoint map* $Q^\dagger$ is an operator defined by

$$\langle\psi|\,Q^\dagger = (Q\,|\psi\rangle)^\dagger$$

for any $|\psi\rangle$.

Equivalently

$$\begin{aligned}\langle\phi|\,Q^\dagger\,|\psi\rangle &= (Q\,\langle\phi|)^\dagger\,|\psi\rangle \\ &= \langle\psi|\,Q\,|\phi\rangle^\dagger\end{aligned}$$

for all $|\psi\rangle, |\phi\rangle$.

Definition. (Hermitian) An operator Q is *Hermitian* or *self-adjoint* if

$$Q^\dagger = Q$$

and such operators are called *observables* because they correspond to physical quantities that can be measured, for example position, momentum, energy.

To define a quantum system, we must specify the *Hamiltonian* whose eigenvalues are the allowed energy levels of the system,

$$H|\chi\rangle = E|\chi\rangle.$$

We have the following key results for any Hermitian operator Q.

Proposition. For a Hermitian operator Q,
(i) All eigenvalues are real;
(ii) Eigenstates with distinct eigenvalues are orthogonal;
(iii) Any state can be written as a linear combination of its eigenstates (i.e its matrix representation is diagonalisable).

For a proof of (iii) we have a proof in IB Linear Algebra. We'll cover a proof of (i) and (ii) here.

Proof. Suppose that $Q|\psi\rangle = \lambda|\psi\rangle$. So taking Hermitian conjugates on both sides we have that

$$\begin{aligned}\langle\psi|Q^\dagger &= \lambda^*\langle\psi| \\ &= \langle\psi|Q.\end{aligned}$$

Hence $\langle\psi|Q|\psi\rangle = \lambda\langle\psi|\psi\rangle = \lambda^*\langle\psi|\psi\rangle$. Since $|\psi\rangle$ is non-zero, it's norm is non-zero hence $\lambda = \lambda^*$.

For the second part let $|n\rangle$ be an eigenstate of Q with eigenvalues $\lambda = q_n$ real. Then we have that

$$\begin{aligned}Q|n\rangle &= q_n|n\rangle \\ Q|m\rangle &= q_m|m\rangle \\ \langle m|Q &= q_m\langle m|\end{aligned}$$

hence $\langle m|Q|n\rangle = q_n\langle m|n\rangle = q_m\langle m|n\rangle$. But we have that $q_m \neq q_n$ so $\langle m|n\rangle = 0$. □

2.2 The Harmonic Oscillator

A one-dimensional harmonic oscillator of mass m and frequency ω is a system with position $\hat{x}$, momentum $\hat{p}$ and Hamilitonian

$$H = \frac{1}{2m}\hat{p}^2 + \frac{1}{2}m\omega^2\hat{x}^2.$$

We seek the energy eigenstates and eigenvalues of H. We do this by introducting *annihilation, creation, and number operators.*

Define

$$a = \left(\frac{m\omega}{2\hbar}\right)^{\frac{1}{2}} \left(\hat{x} + \frac{i\hat{p}}{m\omega}\right)$$
$$a^\dagger = \left(\frac{m\omega}{2\hbar}\right)^{\frac{1}{2}} \left(\hat{x} - \frac{i\hat{p}}{m\omega}\right).$$

Equivalently

$$\hat{x} = \left(\frac{\hbar}{2m\omega}\right)^{\frac{1}{2}} \left(a + a^\dagger\right)$$
$$\hat{p} = \left(\frac{\hbar m\omega}{2}\right)^{\frac{1}{2}} i\left(a^\dagger - a\right).$$

It's easy to check that

$$[\hat{x}, \hat{p}] = i\hbar \iff [a, a^\dagger] = 1.$$

In fact,

$$\begin{aligned} aa^\dagger &= \frac{m\omega}{2\hbar}\hat{x}^2 + \frac{1}{2m\omega\hbar}\hat{p}^2 - \frac{i}{2\hbar}(\hat{x}\hat{p} - \hat{p}\hat{x}) \\ &= \frac{1}{\hbar\omega}H + \frac{1}{2} \end{aligned}$$

but

$$a^\dagger a = \frac{1}{\hbar\omega}H - \frac{1}{2}.$$

This confirms the commutation relation, but we also find that

$$H = \hbar\omega\left(N + \frac{1}{2}\right)$$

with $N = a^\dagger a$, the *number operator* which is clearly Hermitian, so finding eigenvalues and eigenstates of N is equivalent to finding these for H.

Let $|\lambda\rangle$ be any eigenstate of N with eigenvalue λ, so

$$N|\lambda\rangle = \lambda|\lambda\rangle \iff H|\lambda\rangle = E|\lambda\rangle$$

where $E = \hbar\omega\left(\lambda + \frac{1}{2}\right)$. Then assuming that $|\lambda\rangle$ is normalised, write

$$\lambda = \langle\lambda| N |\lambda\rangle = \lambda a^\dagger a|\lambda\rangle = (a|\lambda\rangle)^\dagger (a|\lambda\rangle) = ||a|\lambda\rangle||^2$$

we could have just seen this directly, since N is clearly positive semi-definite from its construction as the outer product of an operator. So we have that $\lambda \geq 0$ and $\lambda = 0 \iff a|\lambda\rangle = 0$.

Now we consider

$$\begin{aligned} N(a^\dagger|\lambda\rangle) &= ([N, a^\dagger] + a^\dagger N)|\lambda\rangle \\ &= (a^\dagger + a^\dagger\lambda)|\lambda\rangle \\ &= (\lambda + 1)(a^\dagger|\lambda\rangle). \end{aligned}$$

Similarly we have that

$$\begin{aligned} N(a\,|\lambda\rangle) = &= ([N,a] + aN)\,|\lambda\rangle \\ &= (-a + a\lambda)\,|\lambda\rangle \\ &= (\lambda - 1)a\,|\lambda\rangle\,. \end{aligned}$$

Here we have used the results for commutators $[N, a^\dagger] = [a^\dagger, a^\dagger] = a^\dagger[a, a^\dagger] = a^\dagger$ and similarily $[N, a] = -a$.

Now the news states with be eigenstates provided their norms are non-zero. We can compute

$$\begin{aligned} ||a\,|\lambda\rangle\,||^2 &= \lambda \\ ||a^\dagger\,|\lambda\rangle\,||^2 &= \langle\lambda|\,aa^\dagger\,|\lambda\rangle = \lambda a^\dagger a + 1\,|\lambda\rangle \\ &= \lambda + 1. \end{aligned}$$

Because of these properties $a^\dagger$ and a are called *creation* (or raising) and *annihilation* (or lowering) operators.

We can't lower below zero, since the eigenvalues are positive so the eigenvalues must be a subset of the positive integers to avoid a contradiction. We can lower and raise all we want, so we can generate any other state with eigenvalues being all of the non-negative integers.